VGA Controller — Architecture Mind Map

vga_controller.v | 640×480 @ 60Hz VGA | 4-bit RGB output

TIMING ENGINE

- H Counter
  h_cnt: 0→799
  (H_TOTAL=800)
- V Counter
  v_cnt: 0→524
  (V_TOTAL=525)
- Active Zone
  h<640, v<480
  → active=1
- H-Sync Pulse
  656≤h<752
  (neg. polarity)
- V-Sync Pulse
  490≤v<492
  (neg. polarity)
- Pixel Clock
  25 MHz (clk25)

OBJECT DETECTION

- Paddle Body
  px≤h<px+pw
  py≤v<py+10
- Paddle Highlight
  Top 2 rows of
  paddle (brighter)
- Ball
  bx≤h<bx+10
  by≤v<by+10
- Border
  h<4 or h≥636
  or v<4 or v≥476
- Game Over Box
  game_over=1
  180≤h<460
  160≤v<320

VGA CONTROLLER

PIXEL PRIORITY ORDER (highest → lowest)

1. Blanking (outside active) → Black
2. Score digits (top-right)
3. Game Over box (border / fill / text)
4. Paddle highlight (top 2px) → Bright Cyan
5. Paddle body → Cyan 6. Ball → White
7. Border → Gray 8. Background → Dark Navy

SCORE DISPLAY

- draw_digit()
  function
- 3 digits:
  H=score/100
  T=score/10
  O=score%10
- Positions:
  560,576,592
  (top-right HUD)
- Scale ×3
  15px wide
  21px tall

FONT SYSTEM (5×7)

- Font ROM
  19 glyphs stored
  in reg [4:0] font
  [0:18][0:6]
- Chars 0–8
  G A M E ⎕ O V E R
  (9 glyphs)
- Chars 9–18
  Digits 0–9
  (10 glyphs)
- Scale Mode
  Score: ×3
  (15×21 px each)
- Scale Mode
  Game Over: ×4 wide
  ×3 tall (20×21 px)
- Bit Extract
  font[id][row]
  [4-col]

INPUT SIGNALS:

- ▸ clk25 (pixel clk)
- ▸ player_x/y/w
- ▸ ball_x/y
- ▸ score[9:0]
- ▸ game_over

- ▸ vga_r[3:0]
- ▸ vga_g[3:0]
- ▸ vga_b[3:0]
- ▸ vga_hs
- ▸ vga_vs

GAME OVER OVERLAY

- draw_gameover()
  function
- Box: 180..460
  × 160..320
  (280×160 px)
- Text centered:
  "GAME OVER"
  (9 chars @ 20px)
- 3px orange
  border around
  the box

COLOR OUTPUT

- Blanking
  RGB = 000
  (outside active)
- Score Digits
  RGB = FF0
  (Yellow)
- Game Over Box
  Border: FA0 (Orange)
  Fill: 102 (D.Purple)
  Text: FFF (White)
- Paddle Highlight
  RGB = 8FF
  (Cyan-hi)
- Paddle Body
  RGB = 0CC
  (Cyan)
- Ball = FFF
  Border = AAA
  BG = 012
  (Dark Navy)